

Step 3: Taylor expansion of the projection map. Since $\|\mathbf{y}_t\| \geq 1/2$ on $\mathcal{E}_t$, the projection map $\Pi(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|$ is C^2 on a neighborhood of the trajectory. Therefore,

$$\Pi(\mathbf{y}_t + \rho \boldsymbol{\zeta}_{t+1}) - \Pi(\mathbf{y}_t) = \rho J_t \boldsymbol{\zeta}_{t+1} + \mathbf{R}_{t+1},$$

where J_t is the Jacobian matrix

$$J_t := D\Pi(\mathbf{y}_t) = \frac{1}{\|\mathbf{y}_t\|} \left(I - \Pi(\mathbf{y}_t) \Pi(\mathbf{y}_t)^\top \right),$$

and the remainder satisfies

$$\|\mathbf{R}_{t+1}\| \leq C_R \rho^2 \|\boldsymbol{\zeta}_{t+1}\|^2 \leq 4C_R \rho^2$$

for a universal constant $C_R > 0$. Moreover, since $\|\mathbf{y}_t\| \geq 1/2$, we have $\|J_t\| \leq 2$. Substituting this expansion into (16), we obtain

$$\|\Pi(\mathbf{z}_{t+1}) - \boldsymbol{\mu}_t\|^2 = \|\mathbf{a}_t\|^2 + 2\rho \langle \mathbf{a}_t, J_t \boldsymbol{\zeta}_{t+1} \rangle + \Xi_{t+1}, \quad (17)$$

where Ξ_{t+1} collects the higher-order terms:

$$\Xi_{t+1} := 2\rho \langle \mathbf{a}_t, \mathbf{R}_{t+1} \rangle + \|\rho J_t \boldsymbol{\zeta}_{t+1} + \mathbf{R}_{t+1}\|^2.$$

Since $\|\mathbf{a}_t\| \leq 2$, $\|J_t\| \leq 2$, $\|\boldsymbol{\zeta}_{t+1}\| \leq 2$, and $\|\mathbf{R}_{t+1}\| \leq 4C_R \rho^2$, there exists a universal constant $C_1 > 0$ such that

$$|\Xi_{t+1}| \leq C_1 \rho^2.$$

Let $\mathbb{1}_{\mathcal{E}_t}$ denote the indicator of $\mathcal{E}_t$ and define

$$\Delta_{t+1} := 2\rho \langle \mathbf{a}_t, J_t \boldsymbol{\zeta}_{t+1} \rangle \mathbb{1}_{\mathcal{E}_t}.$$

Since $\mathbf{a}_t$, J_t , and $\mathbb{1}_{\mathcal{E}_t}$ are $\mathcal{F}_t$ -measurable and $\mathbb{E}[\boldsymbol{\zeta}_{t+1} \mid \mathcal{F}_t] = 0$, $\{\Delta_{t+1}\}$ is a martingale difference sequence with respect to $\{\mathcal{F}_t\}$. Moreover, there exist constants $C_2, C_3 > 0$ such that, almost surely,

$$|\Delta_{t+1}| \leq C_2 \rho, \quad \mathbb{E}[\Delta_{t+1}^2 \mid \mathcal{F}_t] \leq C_3 \rho^2.$$

Hence, by (17), on $\mathcal{E}_t$,

$$\|\Pi(\mathbf{z}_{t+1}) - \boldsymbol{\mu}_t\|^2 \leq \|\mathbf{a}_t\|^2 + \Delta_{t+1} + C_1 \rho^2.$$

Step 4: One-step recursion on the bootstrap region. By Lemma G.8,

$$\|\mathbf{a}_t\|^2 \leq (1 - c_0 \rho)^2 \|\mathbf{e}_t\|^2 \leq (1 - c_0 \rho) \|\mathbf{e}_t\|^2.$$

Combining this with (15), we obtain that on $\mathcal{E}_{t-1}$,

$$\|\mathbf{e}_{t+1}\|^2 \leq (1 + \gamma)(1 - c_0 \rho) \|\mathbf{e}_t\|^2 + (1 + \gamma) C_1 \rho^2 + \left(1 + \frac{1}{\gamma}\right) C_\mu^2 \eta^2 + (1 + \gamma) \Delta_{t+1}.$$

We then choose $\gamma := \frac{c_0 \rho}{4}$. As a result, for ρ_0 sufficiently small, we could simultaneously have that (i) $(1 + \gamma)(1 - c_0 \rho) \leq 1 - c_1 \rho$ for some constant $c_1 > 0$, (ii) $(1 + \gamma) C_1 \rho^2 \leq C_4 \rho^2$, and (iii) $\left(1 + \frac{1}{\gamma}\right) C_\mu^2 \eta^2 \leq C_5 \frac{\eta^2}{\rho}$. As a result of the above notation simplification, we may rewrite the recursion as

$$\|\mathbf{e}_{t+1}\|^2 \leq (1 - c_1 \rho) \|\mathbf{e}_t\|^2 + C_4 \rho^2 + C_5 \frac{\eta^2}{\rho} + \left(1 + \frac{c_0 \rho}{4}\right) \Delta_{t+1}, \quad (18)$$

where $\{\Delta_{t+1}\}$ is a martingale difference sequence satisfying $|\Delta_{t+1}| \leq C_2 \rho$ and $\mathbb{E}[\Delta_{t+1}^2 \mid \mathcal{F}_t] \leq C_3 \rho^2$.

Step 5: Unrolling the recursion. Iterating (18) yields that, on $\mathcal{E}_{t-1}$,

$$\|\mathbf{e}_t\|^2 \leq (1 - c_1 \rho)^t \|\mathbf{e}_0\|^2 + \left(C_4 \rho^2 + C_5 \frac{\eta^2}{\rho}\right) \sum_{j=0}^{t-1} (1 - c_1 \rho)^j + \left(1 + \frac{c_0 \rho}{4}\right) \sum_{s=0}^{t-1} (1 - c_1 \rho)^{t-1-s} \Delta_{s+1}.$$

Since $\sum_{j=0}^{t-1} (1 - c_1 \rho)^j \leq \frac{1}{c_1 \rho}$, we obtain that on $\mathcal{E}_{t-1}$,

$$\|\mathbf{e}_t\|^2 \leq (1 - c_1 \rho)^t \|\mathbf{e}_0\|^2 + C_6 \rho + C_7 \frac{\eta^2}{\rho^2} + \left(1 + \frac{c_0 \rho}{4}\right) \sum_{s=0}^{t-1} (1 - c_1 \rho)^{t-1-s} \Delta_{s+1}. \quad (19)$$

Now, we apply the concentration inequality in Lemma G.9 to the martingale sequence $\{\Delta_{t+1}\}$ (with $c_\star = c_1$). There exists a constant $C_8 > 0$ such that, with probability at least $1 - \delta/2$,

$$\left(1 + \frac{c_0\rho}{4}\right) \sup_{1 \leq t \leq T} \left| \sum_{s=0}^{t-1} (1 - c_1\rho)^{t-1-s} \Delta_{s+1} \right| \leq C_8 \left(\sqrt{\rho \log \frac{2T}{\delta}} + \rho \log \frac{2T}{\delta} \right). \quad (20)$$

We denote the event in (20) as $\mathcal{E}_\#$. Then $\mathbb{P}(\mathcal{E}_\#) \geq 1 - \delta$. Since $\rho \leq 1$ and $\rho \log(2T/\delta) \leq 1$, after enlarging constants we can bound the right-hand side by $C_9 \sqrt{\rho \log \frac{2T}{\delta}}$.

Combining this with (19), we conclude that on the event $\mathcal{E}_\# \cap \mathcal{E}_{t-1}$, for suitable constants $c, C > 0$,

$$\|\mathbf{e}_t\|^2 \leq (1 - c\rho)^t \|\mathbf{e}_0\|^2 + C \left(\sqrt{\rho \log \frac{2T}{\delta}} + \frac{\eta^2}{\rho^2} \right). \quad (21)$$

Step 6: Induction on Bootstrap events. By the condition in Theorem G.2,

$$\|\mathbf{e}_0\|^2 + C \left(\sqrt{\rho \log \frac{2T}{\delta}} + \frac{\eta^2}{\rho^2} \right) \leq \frac{1}{16}. \quad (22)$$

As a result of this condition, $\mathcal{E}_0 = \{\|\mathbf{e}_0\| \leq \frac{1}{4}\}$ is true. Now, we want to show that for any $t \geq 1$, on the event $\mathcal{E}_\# \cap \mathcal{E}_{t-1}$, we must have that the event $\mathcal{E}_t$ is also true. It suffices to show that

$$\|\mathbf{e}_t\|^2 \leq \frac{1}{16},$$

which naturally follows by combining (22) and (21). We then complete the proof by induction. $\square$

G.5 Proof of Lemma G.5

Proof of Lemma G.5. We split this region $\widehat{\mathcal{R}} \Delta \mathcal{R}^\star$ into two disjoint parts: $\mathcal{R}^\star \cap \widehat{\mathcal{R}}^c$ (where the optimal classifier accepts but the estimated one rejects) and $(\mathcal{R}^\star)^c \cap \widehat{\mathcal{R}}$ (where the estimated one accepts but the optimal rejects). It then follows that

$$\begin{aligned} & \int_{(\mathcal{R}^\star \cap \widehat{\mathcal{R}}^c) \cup ((\mathcal{R}^\star)^c \cap \widehat{\mathcal{R}})} \left| \frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^\star} \right| d\mathbb{P}_0(x) \\ &= \int_{\mathcal{R}^\star \cap \widehat{\mathcal{R}}^c} \left| \frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^\star} \right| d\mathbb{P}_0(x) + \int_{(\mathcal{R}^\star)^c \cap \widehat{\mathcal{R}}} \left| \frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^\star} \right| d\mathbb{P}_0(x) \\ &= \int_{\mathcal{R}^\star \cap \widehat{\mathcal{R}}^c} \left(e^{\tau_\alpha^\star} - \frac{p_1(x)}{p_0(x)} \right) d\mathbb{P}_0(x) + \int_{(\mathcal{R}^\star)^c \cap \widehat{\mathcal{R}}} \left(\frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^\star} \right) d\mathbb{P}_0(x). \end{aligned}$$

Note that

$$\begin{aligned} & \int_{\mathcal{R}^\star} \left(e^{\tau_\alpha^\star} - \frac{p_1(x)}{p_0(x)} \right) d\mathbb{P}_0(x) + \int_{\widehat{\mathcal{R}}} \left(\frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^\star} \right) d\mathbb{P}_0(x) \\ &= \underbrace{\int_{\mathcal{R}^\star \cap \widehat{\mathcal{R}}} \left(e^{\tau_\alpha^\star} - \frac{p_1(x)}{p_0(x)} + \frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^\star} \right) d\mathbb{P}_0(x)}_{=0} \\ & \quad + \int_{\mathcal{R}^\star \cap \widehat{\mathcal{R}}^c} \left(e^{\tau_\alpha^\star} - \frac{p_1(x)}{p_0(x)} \right) d\mathbb{P}_0(x) + \int_{(\mathcal{R}^\star)^c \cap \widehat{\mathcal{R}}} \left(\frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^\star} \right) d\mathbb{P}_0(x). \end{aligned}$$

Using the property that $\int_A \frac{p_1(x)}{p_0(x)} d\mathbb{P}_0(x) = \int_A d\mathbb{P}_1(x) = \mathbb{P}_1(A)$ for any measurable set A , we have

$$\begin{aligned} & \int_{(\mathcal{R}^\star \cap \widehat{\mathcal{R}}^c) \cup ((\mathcal{R}^\star)^c \cap \widehat{\mathcal{R}})} \left| \frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^\star} \right| d\mathbb{P}_0(x) \\ &= \int_{\mathcal{R}^\star} \left(e^{\tau_\alpha^\star} - \frac{p_1(x)}{p_0(x)} \right) d\mathbb{P}_0(x) + \int_{\widehat{\mathcal{R}}} \left(\frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^\star} \right) d\mathbb{P}_0(x) \\ &= \mathbb{P}_1(\widehat{\mathcal{R}}) - \mathbb{P}_1(\mathcal{R}^\star) + e^{\tau_\alpha^\star} \left(\mathbb{P}_0(\mathcal{R}^\star) - \mathbb{P}_0(\widehat{\mathcal{R}}) \right), \end{aligned}$$

which, together with the fact that $\mathbb{P}_0((\mathcal{R}^\star)^c) = \alpha$, implies that $\mathbb{P}_0(\mathcal{R}^\star) = 1 - \alpha$. Recalling $\mathbb{P}_0(\widehat{\mathcal{R}}) = 1 - \mathbb{P}_0(\widehat{\mathcal{R}}^c)$, we obtain $\mathbb{P}_0(\mathcal{R}^\star) - \mathbb{P}_0(\widehat{\mathcal{R}}) = \mathbb{P}_0(\widehat{\mathcal{R}}^c) - \alpha$. We then complete the proof. $\square$

G.6 Proof of Lemma G.6

Proof of Lemma G.6. By the uniform error bound, we have

$$\{\mathcal{S}(X) > \hat{\tau}_\alpha + r(\delta, \rho, \eta)\} \subseteq \{\hat{\mathcal{S}}(X) > \hat{\tau}_\alpha\} \subseteq \{\mathcal{S}(X) > \hat{\tau}_\alpha - r(\delta, \rho, \eta)\},$$

which implies

$$\mathbb{P}_0(\mathcal{S}(X) > \hat{\tau}_\alpha + r(\delta, \rho, \eta)) \leq \mathbb{P}_0(\hat{\mathcal{R}}^c) \leq \mathbb{P}_0(\mathcal{S}(X) > \hat{\tau}_\alpha - r(\delta, \rho, \eta)).$$

Since $\alpha = \mathbb{P}_0(\mathcal{S}(X) > \tau_\alpha^*)$, it follows that $\mathbb{P}_0(\mathcal{S}(X) \in [\tau_\alpha^*, \hat{\tau}_\alpha - r(\delta, \rho, \eta)]) \leq \Delta_{n_2}(\delta)$ whenever $\hat{\tau}_\alpha - r(\delta, \rho, \eta) \geq \tau_\alpha^*$, and similarly $\mathbb{P}_0(\mathcal{S}(X) \in [\hat{\tau}_\alpha + r(\delta, \rho, \eta), \tau_\alpha^*]) \leq \Delta_{n_2}(\delta)$ whenever $\hat{\tau}_\alpha + r(\delta, \rho, \eta) \leq \tau_\alpha^*$.

Now let

$$u_\delta := \left(\frac{\Delta_{n_2}(\delta)}{c_0} \right)^{1/\gamma}.$$

Assume $u_\delta < \varepsilon_0$. By the one-sided lower bounds in Assumption 3, for every $\varepsilon \in [0, \varepsilon_0]$,

$$\mathbb{P}_0(\tau_\alpha^* < \mathcal{S}(X) \leq \tau_\alpha^* + \varepsilon) \geq c_0 \varepsilon^\gamma \quad \text{and} \quad \mathbb{P}_0(\tau_\alpha^* - \varepsilon \leq \mathcal{S}(X) < \tau_\alpha^*) \geq c_0 \varepsilon^\gamma.$$

Suppose first that $\hat{\tau}_\alpha - r(\delta, \rho, \eta) > \tau_\alpha^* + u_\delta$. Then one can choose $\varepsilon \in (u_\delta, \varepsilon_0]$ such that

$$\tau_\alpha^* + \varepsilon < \hat{\tau}_\alpha - r(\delta, \rho, \eta).$$

The right-sided lower bound therefore gives

$$\mathbb{P}_0(\tau_\alpha^* < \mathcal{S}(X) \leq \tau_\alpha^* + \varepsilon) > \Delta_{n_2}(\delta),$$

contradicting the preceding upper bound on the probability mass between τ_α^* and $\hat{\tau}_\alpha - r(\delta, \rho, \eta)$. The case $\hat{\tau}_\alpha + r(\delta, \rho, \eta) < \tau_\alpha^* - u_\delta$ is analogous, using the left-sided lower bound. Hence,

$$|\hat{\tau}_\alpha - \tau_\alpha^*| \leq r(\delta, \rho, \eta) + u_\delta.$$

Next, consider any $x \in \hat{\mathcal{R}} \Delta \mathcal{R}^*$.

(i) If $x \in (\mathcal{R}^*)^c \cap \hat{\mathcal{R}}$, then $\mathcal{S}(x) > \tau_\alpha^*$ and $\hat{\mathcal{S}}(x) \leq \hat{\tau}_\alpha$. Using the uniform score bound gives

$$\mathcal{S}(x) \leq \hat{\tau}_\alpha + r(\delta, \rho, \eta) \leq \tau_\alpha^* + 2r(\delta, \rho, \eta) + \left(\frac{\Delta_{n_2}(\delta)}{c_0} \right)^{1/\gamma}.$$

(ii) If $x \in \mathcal{R}^* \cap \hat{\mathcal{R}}^c$, then $\mathcal{S}(x) \leq \tau_\alpha^*$ and $\hat{\mathcal{S}}(x) > \hat{\tau}_\alpha$, and hence

$$\tau_\alpha^* - 2r(\delta, \rho, \eta) - \left(\frac{\Delta_{n_2}(\delta)}{c_0} \right)^{1/\gamma} < \mathcal{S}(x) \leq \tau_\alpha^*.$$

Combining the two cases, we complete the proof. $\square$

G.7 Useful Lemmas

Lemma G.7. Let $m > 0$. For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ satisfying $\|\mathbf{x}\| \geq m$ and $\|\mathbf{y}\| \geq m$,

$$\|\Pi(\mathbf{x}) - \Pi(\mathbf{y})\| \leq \frac{2}{m} \|\mathbf{x} - \mathbf{y}\|.$$

Proof of Lemma G.7. Using $\Pi(\mathbf{x}) - \Pi(\mathbf{y}) = \frac{\mathbf{x} - \mathbf{y}}{\|\mathbf{x}\|} + \mathbf{y} \left(\frac{1}{\|\mathbf{x}\|} - \frac{1}{\|\mathbf{y}\|} \right)$, we obtain

$$\|\Pi(\mathbf{x}) - \Pi(\mathbf{y})\| \leq \frac{\|\mathbf{x} - \mathbf{y}\|}{\|\mathbf{x}\|} + \frac{\|\mathbf{y}\| - \|\mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{2}{m} \|\mathbf{x} - \mathbf{y}\|.$$

$\square$